

DSC 478 Final Project Report

Understanding Public Sentiment and Opinion on the Israel-Palestine Conflict Through Reddit Discourse.

Project Type: Data Analysis

Team Members [student ID]: Mohammed Musaddiq Vavartar [2110431]

Course: DSC 478 - Programming Machine Learning Applications

Instructor: Bamshad Mobasher

Demo video: <https://www.youtube.com/watch?v=yxqX6NZYPPo>

1. Introduction & Objective

1.1 Project Goal and Research Question:

This data analysis project is aimed to apply knowledge discovery in Databases based on a large real world dataset of Reddit comments about the Israel-Palestine conflict. The main goal is to understand what people genuinely think and feel about this conflict by analyzing over one million reddit comments using sentiment analysis, emotion detection, unsupervised learning and supervised classification.

The project's clear primary goal is based on the research question:

What are peoples genuine stances, opinions and emotional sentiments toward the Israel-Palestine conflict as expressed on Reddit, Are they postive, negative, neutral, driven by anger, disgust, surprise, joy, sadness, neutral or something else?

A secondary objective explores whether these stances and emotions shifted between two distinct time periods, pre 2024 and post 2024,

1.2 Why Reddit Data?

Reddit is chosen as a data source for three reasons. First, Reddit comments are user's raw emotional expression (unlike filtered news headlines). Second, the dataset is large enough for robust machine learning. Third, subreddit structure makes topic filtering possible (r/Israel, r/Palestine, r/AskMiddleEast) to keep the data super relevant and remove off topic noise

1.3 Pre/Post 2024 Split:

A split timeline was chosen because, as per the mainstream media, it marks the major incident of Hamas attack on Israel, which triggered this major conflict. The data shows a clear initial and after moment in public discourse. The pre 2024 after filter has 414,710 comments (Oct 2023 - Dec 2023) comments displaying the immediate aftermath of the attack. The post 2024 after filter contains 622,108 comments shows the extended conflict comments. This split main purpose is to notice if there are changes in response to a major real world event.

1.4 Overview of Methods:

Section 1 Intial Setup: Google drive mount[1.1], Gdrive project directory [1.2], initial libraries installation[1.3] and downloading dataset from kaggle using Kaggle API [1.4][1.5], Dataset insights [1.6][1.7]

Section 2 Data loading and Filtering: Chunk wise 6 step filter for main dataset [2.1][2.2], Remove duplicates [2.3], three step filter for legacy dataset [2.4], producing two clean datasets totaling 1.036,818 comments [2.5], Showing top 5 comments[2.6].

Section 3 Exploratory Data Analysis: Dataset shape [3.1], Comment scores [3.2], top active subreddits [3.3], comments length [3.4], top users [3.5], upvote ratios [3.6] are [analyzed and visualized for both datasets.](#)

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Section 4 Sentiment and Emotion Detection: Layer 1 uses VADER and TextBlob with custom sarcasm rules to label each comments are +ve, _ve or sarcasm [4.1.1 - 4.1.4]. Layer 2 uses the [HuggingFace distrillroberta model](#) to detect 7 emotions (anger, disgust, fear, joy, neutral, sadness, surprise) [4.2.1 - 4.2.9].

Section 5 Unsupervised Learning: LDA topic modeling found 7 topics and K means clustering found 5 clusters, Both are applied on TF-IDF vectors and compared across pre and post 2024. [5.1.1 - 5.1.6]. Using TF-IDF matrix & Testing k using elbow method for K means clustering [5.2.1 - 5.2.4]

Section 6 Supervised Learning: Five classifier were trained on both VADER labels [6.1.1 - 6.1.8] and hugging face emotion labels [6.2.1 - 6.2.5].

Section 7 Conclusion: Findings, VADER vs HuggingFace comparison and limitations are talked.

Note: The code implementation mentioned in this report is related to cells labeled with numbers in the associated notebook file, for example: [4.1.1] cell marks the start of VADER sentiment analysis

2. Dataset & preprocessing

2.1 Dataset Description :

The dataset is taken from Kaggle "Reddit on Israel-Palestine Daily Updated" by Asaniczka.

Source: <https://www.kaggle.com/datasets/asaniczka/reddit-on-israel-palestine-daily-updated>

The current dataset includes data for October 2023 to June 2026 with total columns of 24 which include comment body, score, subreddit, author, upvote ratio and user karma.

2.2 Data Filtering :

A 6 step filtering to both datasets to ensure data quality and topic relevance,

Filter	What it removes	why?
Bot removal	User_is_verified = False AutoModerator posts	Make sure all comments are from real human
Topic relevance	Keep only 3-5 subreddits	Focus on conflict specific discussion
Word count	Remove comments under 10 or over 500	Under 10 lack meaning and over 500 is mixed opinions
Zero engagement	Remove score = 0 comments	No social validation
Duplicate user post pairs	Keep only highest scored comment per user per post	Reduces bias from repeated content by same user
Date check	pre/post 2024 split	Separation for dataset

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2.3 Final Dataset Statistics:

After filtering, comments were reduced to two clean datasets:

Dataset	-	# comments	-	columns
Pre 2024	-	414,710	-	5
Post 2025	-	622,108	-	24
Total	-	1,036,818		

2.4 Exploratory Data Analysis:

Scores for Comments Mean comment score before 2024 was 5.18 and 11.53 with median 2 and 3 for after 2024 respectively. This shows higher engagement after October 7th. The highest scored post 2024 was upvoted 3,023 times.

Subreddits of Posts The top subreddit both before and after 2024 was r/IsraelPalestine with shares 67.8% and 56.3% respectively. There is also a large increase in r/Palestine from 15.2% to 31.5% after 2024.

Lengths of Comments Comment lengths were relatively short. Before 2024, 66% had less than 50 words. Average word count before 2024 and after was 53.6 and 54.4 words almost the same showing comment length remained relatively unchanged despite the conflict.

Users' Activity No particular user stood out dominating the discussion. JosephL_55 contributed the largest number of comments which was 1,965 in a total of 622,108 0.3% of all post 2024 comments.

Upvote Ratio of Comments Mean Upvote ratio after 2024 was 0.760. 38% of the comments came from posts with upvote ratios 0.90 whereas 13.9% were on controversial posts

3. Sentiment & Emotion Detection

3.1 Design Decision | Two Layer:

The two-layered approach has been taken to answer the research question, as the single sentiment analysis tool approach cannot give a complete answer to the research question. While VADER can tell us whether a comment is positive or negative, it doesn't provide us information about the exact emotion behind it. The HuggingFace emotion classifier model addresses this concern. Each layer of sentiment classification has been carried out on both datasets independently of each other before applying any machine learning algorithms in Section 5.

3.2 Layer 1: VADER + TextBlob + Sarcasm Rules:

SentimentIntensityAnalyzer [4.1.1] from *vaderSentiment* was created and used on each comment with *polarity_scores()* [4.1.1], which gives back a score ranging from -1 to +1. *TextBlob* is used on the comments separately with *.sentiment.polarity* [4.1.1]. The *vader_score* that finally gets stored with each comment is calculated as $\text{round}((\text{vader_score} + \text{textblob_score})/2, 4)$.

To label, created an *is_sarcastic()* [4.1.1] function using regular expressions through *re.search()*. Two list objects were created where *sarcasm_words* included 17 trigger words like *great job*, *i love how*, *how wonderful while* *conflict_words* included 30 violence-related words like *killing*, *bombing*, *genocide*. If a trigger phrase was followed by any violence-related word then the comment was labeled sarcastic.

In *get_combined_label()* [4.1.1], sarcasm has priority over everything else at first. If not a sarcastic comment, the value from VADER compound would determine whether it was positive or negative, with positive score 0.05, negative

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score -0.05 , or if in between, then sarcasm/statement. All comments went through these two functions using *pandas* *apply()*, creating *vader_score* [4.1.1] and *vader_label* columns for both dfs.

Result: Pre 2024 : 53.6% negative, 35.5% positive, 11.1% sarcasm. Post 2024: 51.9% negative, 36.1% positive, 12.0% sarcasm.

3.3 Layer 2: HuggingFace Emotion Detection:

Model was imported using *transformers.pipeline* [4.2.2] with *task='text-classification'* and *top_k=1*. Detection of GPU was carried out using *torch.cuda.is_available()* [4.2.3], which detected device 0 for GPU use while -1 was used for CPU.

For each comment, a *clean_text()* function was applied before model inference: text was limited to 512 tokens (the maximum number the model can accept), converted to ASCII with *.encode('ascii', 'ignore').decode('ascii')*, and null values were substituted with "no comment".

The inference took place within *run_emotion_model()* [4.2.4] where the model outputted *r[0]['label']* and *r[0]['score']* for every batch of 128 comments in a *tqdm* loop. In cases where inference in a batch was not possible because of CUDA problems, the other approach of inferring emotion per comment was used. Every 50,000 row checkpoint was saved using *df.to_csv()*. Both *emotion label* and *emotion score* columns were included in both dataframe exports.

Result: pre 2024: neutral 39.7%, anger 23.3%, disgust 16.5%. Post 2024: neutral 41.2%, anger 20.3%, disgust 18.7% showing disgust grew post October 7th.

4. Unsupervised Learning

4.1 Text preprocessing:

Before going into LDA and K-Means, a *clean_for_nlp()* [5.1.1] function was used on both datasets via *pandas* *.apply()*. This involved lowering the case of the text, removal of URLs by *re.sub(r'http\S+|www\S+', '', text)*, removal of non-alphabetical characters via *re.sub(r'^a-zA-Z\s', '', text)* [5.1.1], tokenization via *text.split()*, removing words shorter than three characters, removal of NLTK stopwords and additionally 37 more stopwords, which include words related to conflict (*israel*, *palestine*, *hamas*, *gaza*) and occur in every comment without any discrimination toward topic modeling, and Lemmatization via *WordNetLemmatizer*.

4.2 Vectorization:

Matrices for both algorithms has been created based on *clean_text*. *TfidfVectorizer* has been employed for K-Means where *max_features* is set to 10000, *min_df* equals 5, *max_df* equals 0.90, *ngram_range* equals (1,2) [5.1.2], resulting in a matrix of shape (414710, 10000) for pre 2024. For LDA, a *CountVectorizer* has been used with all these parameters the same. Fitted vectorizer on pre 2024 and used *.transform()* on post 2024 to keep same vocab.

4.3 LDA Topic Modeling:

The *LatentDirichletAllocation* class in sklearn was used. For testing different k values, a 50,000 sample set was created using *np.random.choice()* [5.1.3]. K values of 5 through 10 were tested with *learning_method = 'batch'* [5.1.3] and *max_iter = 10*. The log likelihoods were calculated using *.score()* and the highest value, k=7 (-8,996,623.5), was chosen. [\[lda_optimal_K_output_image\]](#) [\[lda_topics_output_image\]](#)

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The final model had parameters `learning_method = 'online'` [5.1.4], `batch_size = 4096`, `n_jobs = -1`, and `max_iter = 10`. This model was trained with the whole 414,710 pre 2024 count matrix. `.transform()` was then run on the two datasets resulting in a (414710, 7) probabilities matrix. Using `argmax(axis=1)+1`, the dominant topic number for each comment was found. Each topic number was labeled by looking at the top 15 words for each using `topic.argsort()[:16:-1]`. The 2nd topic was found to be noise created by bots and mods.

Result: Media and propaganda was the largest topic at 22% pre 2024. Religion and Identity grew from 14.2% to 18.0% post 2024.

4.4 K Means Clustering:

`Minibatchkmeans` is chosen over `KMeans` as it operates on batches and hence much faster for processing one million comments. Elbow method is used for testing `k` from 2 to 10 using `batch_size = 4096` and `n_init` of 3. Inertia became constant for `k=5` hence we chose 5 clusters. [\[kmeans_elbow_output_image\]](#) [\[kmeans_cluster_image\]](#)

In the final clustering `n_clusters = 5` [5.2.1], `batch_size = 4096`, and `n_init = 10`. We fitted on matrix only before 2024. Function `.predict()` is used on both datasets to find cluster assignments. The top 15 words of each cluster are found based on `km_model.cluster_centers_[cluster_idx]` [5.2.2] using `argsort()[:16:-1]`. Bot noise cluster 3 was omitted.

Result: Military Violence cluster was the most dominating in the beginning having 60.9% before 2024 rising to 64.9% after 2024.

5. Supervised Learning

5.1 Data Preparation:

From section 4, the preprocessing of text resulted in the addition of the `clean_text` feature. The datasets used for VADER labels were first filtered via `.dropna(clean_text)` and `.dropna(vader_label)` [6.1.1]. The `.train_test_split` with `test_size=0.2`, `random_state=42` and `stratify=vader_label` yielded 331,768 training and 82,942 testing samples respectively for the pre 2024 dataset. A separate `TfidfVectorizer` [6.1.3] with `max_features=10000`, `min_df=5`, `max_df=0.90`, `ngram_range=(1,2)` and `sublinear_tf=True` was fitted on training data only and The other dataset, post 2024 had its own vectorizer which was fitted independently. The `.fit_transform()` was done on the training set and `.transform()` on the test set to prevent data leakage.

The `add_numerical_features()` [6.2.1] function was coded in order to add the 4 numerical features namely `vader_score`, `emotion_score`, `word_count` and score together through `np.hstack()` and adding it to TF-IDF via `scipy.sparse.hstack()`. This results in the combination of TF-IDF and numerical features yielding an array shape of (331768, 1000

5.2 Model Training - VADER Labels:

Five models are trained using reusable functions `ComplementNB` [6.1.3], `LogisticRegression(max_iter=1000, class_weight='balanced')`, `SGDClassifier(loss='modified_huber', class_weight='balanced', max_iter=300)`, `LinearSVC(class_weight='balanced', max_iter=2000)`, [6.1.3] and `RandomForestClassifier(n_estimators=100, class_weight='balanced')` were used as machine learning models to fit the data, where `class_weight='balanced'` parameter was used on all of them to deal with imbalanced data. StratifiedKFold cross-validation technique was performed on all machine learning models except for Random Forest which is computationally expensive, and accuracy score, precision score, recall score, and `f1_score` where `average='weighted'` were calculated.

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GridSearchCV [6.1.4] was applied to the (best) SVM model with the *parameter* space of $\{C: [0.01, 0.1, 0.5, 1.0, 2.0, 5.0, 10.0]\}$, and $cv=5$. **Best $C=0.1$** gave test accuracy of 78.64% and F1 of 0.7912 on pre 2024. Best $C=0.1$ on post 2024 gave accuracy 78.80% and F1 of 0.7923. [\[comparison_output\]](#) [\[confusion_matrix\]](#)

5.3 Model Training - HuggingFace Emotion Labels:

A **LinearSVC** wrapped by **CalibratedClassifierCV [6.2.1]** was employed for producing probability scores required by the **VotingClassifier** for 7-class emotion classification.

Soft voting ensemble SVM+LR+SGD was formed by **VotingClassifier(voting='soft')**. [6.2.1] The same **train_and_evaluate_emo()** method was utilized with the same CV and metric settings. **ComplementNB** was applied with TF-IDF features as it doesn't work with negative values in a feature set.

[Results:](#) SVM turned out to be the most accurate model (62.25% before 2024, 62.18% after 2024).

5.4 Feature Importance:

SVM Coefficients were generated from **best_svm_pre.coef_[i]** [6.1.6], by each class. The top 20 words were obtained using **np.argsort(coefs)[-20:][::-1]** [6.1.6] and mapped with feature names using **tfidf_v.get_feature_names_out()** [6.1.6]. In case of HuggingFace models, **LinearSVC** used underneath the calibrated model was **svm_clean_pre.calibrated_classifiers_[0].estimator**. [Pre vs Post comparison bar plots](#) were generated for all sentiments classes and all 7 emotions classes. [\[feature_importance_comparison_output\]](#)

Key Observations: Negative Sentiment related words such as 'killed', 'terrorist' and 'kill' remained unchanged before and after 2024. Disgust words became more important after 2024 along with some additional words such as 'dehumanizing'. Sarcasm became more

Conclusion:

In this project, the complete KDD process was done on over one million Reddit posts about Israel-Palestine. The findings have provided an answer to the posed research question discussion on this subject in Reddit is highly negative in tone, and the emotion that prevails throughout both intervals is anger. The most dramatic change observed in the aftermath of the October 7th incident, however, has not been associated with the negativity of the content, but with its emotionality people have gone from being angry to feeling morally outraged as the events unfolded. Media & Propaganda turned out to be the topic that generated most activity, showing that the information battle had been just as fierce as the physical one. SVM with TF-IDF features was the most effective classifier for both 3-class sentiment and 7-class emotion tasks.

Limitations:

The issue with VADER is that it incorrectly classifies sarcasm that uses positive words in conflict contexts. HuggingFace model j-hartmann/emotion-english-distilroberta-base was trained on a general English dataset and not a dataset specifically for conflict text.

The HuggingFace model was trained on a balanced subsample of almost 20,000 observations from 6 different datasets, which gives it an accuracy of 66% on its own test set as compared to a random chance baseline of 14%. Our classifiers, using the labels generated by this model, got an accuracy score of 62.25%, which is slightly less

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than the accuracy score of the model itself. It makes sense that this would be the case since the classifiers were trained using the labels generated by a model that isn't perfect. Any mistakes that the HuggingFace model makes while labeling the data become noise in classifier training, putting us at a ceiling of 66%.

Methods and Tools

Phase	Method	Tool
Data Loading & Filtering	6-step pipeline, chunked reading 100k rows at a time	pandas, os
EDA	Score distributions, subreddit analysis, comment length, user activity, upvote ratio	pandas, matplotlib
Sentiment Analysis	VADER + TextBlob averaged score + custom sarcasm pattern matching	vaderSentiment, TextBlob, re
Emotion Detection	Pretrained transformer inference on GPU, batch size 128, Drive checkpoints	HuggingFace transformers, PyTorch
Text Preprocessing	URL removal, stopword removal, lemmatization, domain stopwords	NLTK, re
Vectorization	TF-IDF (10k features, bigrams, sublinear_tf) + CountVectorizer for LDA	sklearn
Topic Modeling	LDA, log-likelihood k search on 20k sample, optimal k=7	sklearn
Clustering	MiniBatchKMeans, elbow method k=2-10, optimal k=5	sklearn
Supervised Learning	5 classifiers, StratifiedKFold 5-fold CV, GridSearchCV on SVM	sklearn

Note: All hyperlinks to figures in this report refer to images at full resolution saved on the Google Drive. A full collection of all output images are accessible from: [\[HERE\]](#)

Note: Trained models are available at : [\[models_joblib_files\]](#)

Processed dataset pre 2024 : [\[pre2024csv\]](#)

Processed dataset post 2024 : [\[post2024csv\]](#)

All Data csv files are here: [\[Data_folder\]](#)

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